BrickHacks Dashboard

Weather-Normalized Yield Intelligence

Crop yield does not respond to weather in simple ways. Our platform separates expected weather-driven yield from unexplained deviations, helping identify stress impacts, resilience signals, and hidden performance drivers across counties and seasons.

What Drives Yield Variation?

Not all weather variables impact crops equally. Our feature-ranking engine highlights the environmental factors most strongly associated with yield changes, such as heat stress, rainfall distribution, and accumulated growing degree days.

This enables stakeholders to focus on actionable climate risks rather than broad annual averages.

In this data analysis, we analyzed five key factors:

- Precipitation
- Snow Depth
- Snowfall
- Maximum temperature
- Minimum temperature

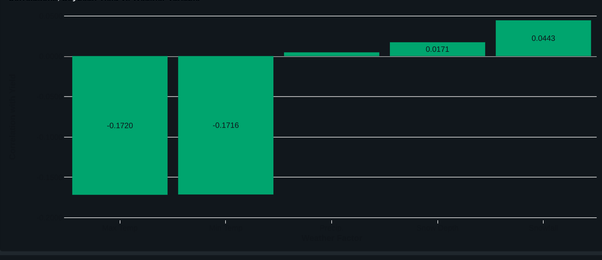
Methodology

We aggregated annual crop yield data for both corn and soybeans throughout the United States, divided down to each individual county. After this, we combined it with weather data obtained from different sensors across the United States, determining sensor location via a county calculation with longitude and latitude.

After the data was cleansed and combined, we calculated Pearson's Correlation Coefficient Scores for all variables and both corn and soybeans.

Correlations, Corn Yield vs Weather Variable

Correlations, Soybean Yield vs Weather Variable



We found the weather variables with higher correlations are *Temp Max* and *Temp Min*.

Where Yield Defied Expectations

We compare actual yield to weather-predicted yield and identify counties that significantly overperformed or underperformed relative to environmental conditions. Overperformance may signal resilience, strong management practices, or favorable microclimates. Underperformance may indicate stress, disease, soil constraints, or unobserved shocks. These anomalies provide starting points for deeper agronomic investigation.

These datapoints highlight unique U.S. counties identified by our model as having exceptional or challenging crop yields. Each county is classified as:

- **Resilient:** Yield was unusually high (top 10% of historical performance)
- **Stress-Impacted:** Yield was unusually low (bottom 10%)
- **Normal:** Yield was within the expected range

All counts and charts below reflect the number of unique counties in each category, not total records. Use this dashboard to quickly spot regions of opportunity or concern.

Resilient County-Years

3473

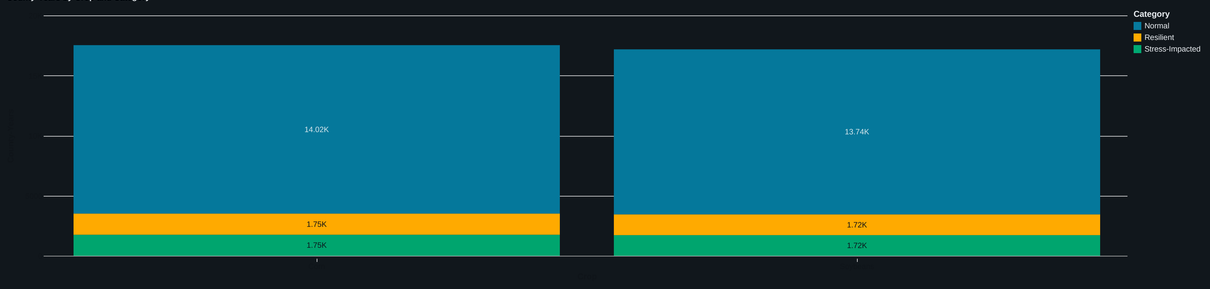
Stress-Impacted County-Years

3474

Normal County-Years

27.77K

County-Years by Crop and Category



Modeling Approach

We train crop-specific models using historical yield and weather data from 2010–2024 to establish a weather-driven yield expectation for each county and season. For each observation, the model producess: Predicted Yield (based on weather conditions), Actual Yield, Residual = Actual – Predicted

The residual represents the Weather-Adjusted Yield Signal, identifying where performance exceeded or underperformed expectations.

Model performance:

RMSE: 18.551 bu/acre

R2 = 0.868

This indicates that weather explains the majority of yield variability while preserving interpretable deviation signals.

Feature	Importance
Total Snow Depth	0.189
Total Precipitation	0.180
Total Snowfall	0.175
Total Max Temperature	0.174
Observation Count	0.169
Total Min Temperature	0.113

Case Study 1 — 2014 Corn (Extreme Outperformance):

Metric	Value
Actual Yield	220.2
Predicted Yield	138.55
Residual	+81.65

Interpretation: Despite moderate modeled weather support, the region dramatically outperformed expectations.Given that snow depth, snowfall, and precipitation are strong drivers, the large accumulated snowpack (5,143) and snowfall (2,261) likely contributed to strong early-season soil moisture reserves. This suggests: Exceptional moisture retention, Possible superior agronomic practices, Potential irrigation or hybrid advantage

Case Study 2:

Metric	Value
Actual Yield	223.1
Predicted Yield	147.07
Residual	+76.03

Interpretation: Precipitation levels were significantly lower (9,445) compared to prior high-moisture years.The model expected suppressed yields due to limited moisture accumulation, yet yields remained strong. This indicates: Structural drought resilience, Possible irrigation coverage, Improved hybrid genetics, Soil health advantagesThis is a prime example of structural performance beyond weather conditions.

Yield variability is primarily driven by:

Seasonal moisture accumulation (precipitation & snowpack), Temperature extremes, Seasonal climate persistence (observation count as proxy for weather intensity).

This confirms agronomic realism: moisture + temperature dominate yield outcomes.

Business Value: Weather-Normalized Yield Insight

Raw yield numbers can be misleading. A strong year may reflect favorable weather, while a weak year may be driven by drought or heat stress beyond anyone's control.
Our system estimates the yield expected under given weather conditions and measures the deviation from that baseline. This weather-normalized signal separates environmental impact from true land or management performance.

For Farmland Investors
Historical averages don't reveal whether land is intrinsically productive or simply benefited from good weather.
By identifying consistent overperformance or underperformance relative to weather conditions, our platform helps investors:
Detect resilient, high-quality landAvoid overpaying for weather-inflated yield historyAssess long-term climate risk exposureThis supports smarter acquisition and valuation decisions.

For Farmers
Year-to-year yield comparisons are often unfair due to changing weather conditions.
Weather-normalized benchmarking allows farmers to:
Evaluate performance more fairly across seasonsDistinguish weather-driven losses from operational issuesDemonstrate resilience and management strength

Core Metric
Yield Residual = Actual Yield – Weather-Predicted Yield
Positive residual → OutperformanceNegative residual → UnderperformanceThis simple metric transforms raw historical data into actionable performance intelligence.